KPI 1: Age Group vs Cardiovascular Disease

Insight:

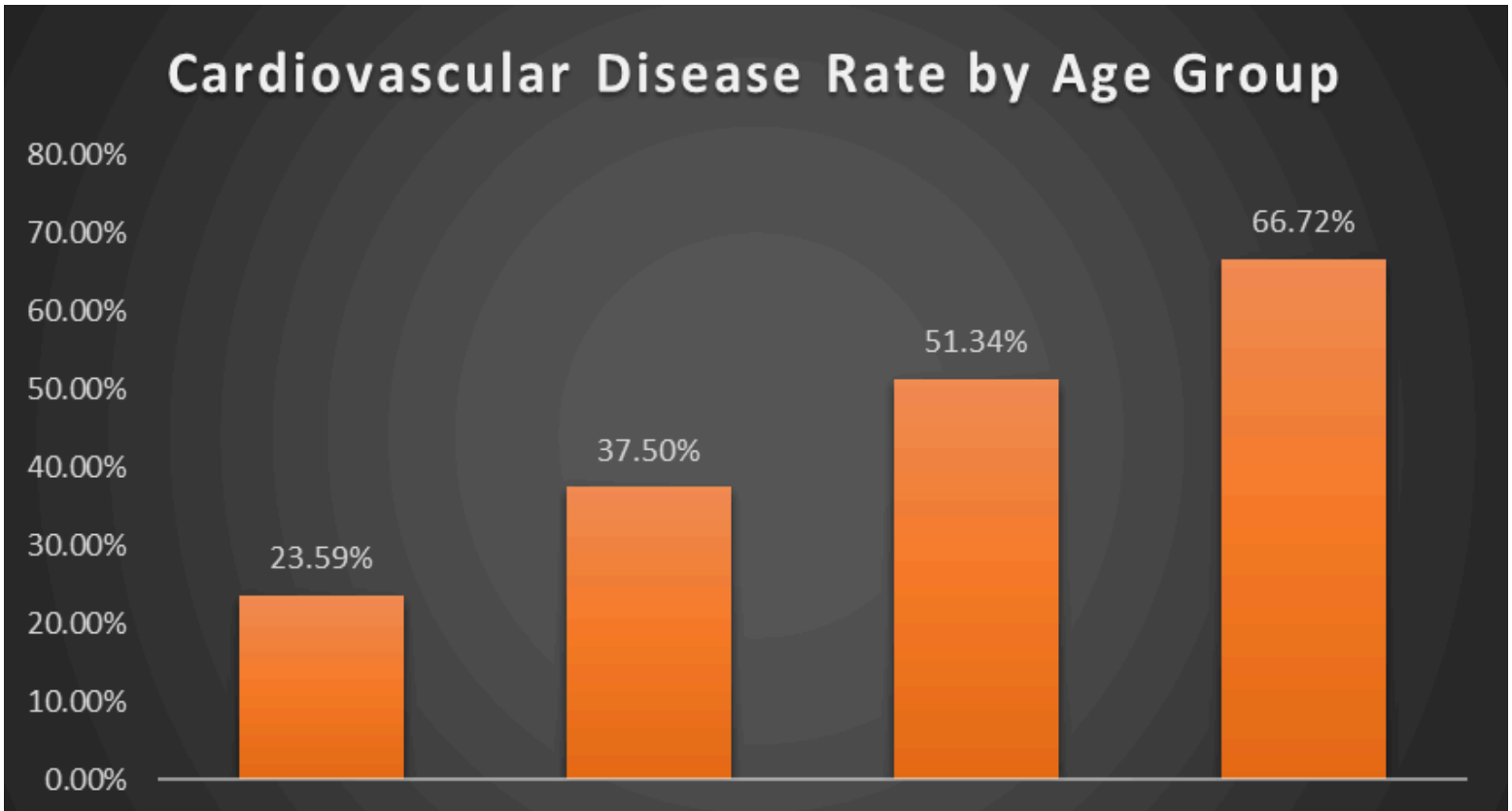
Cardiovascular disease risk increases significantly with age. The prevalence rises from 23.6% in individuals in their 30s to 66.7% in those in their 60s.

Interpretation:

This indicates a strong positive relationship between age and heart disease. As individuals age, physiological changes such as arterial stiffness and increased blood pressure contribute to higher risk.

Business/Healthcare Impact:

Older populations (50s and 60s) should be prioritized for preventive screening and early intervention programs.



KPI 2: Gender vs Cardiovascular Disease

Insight:

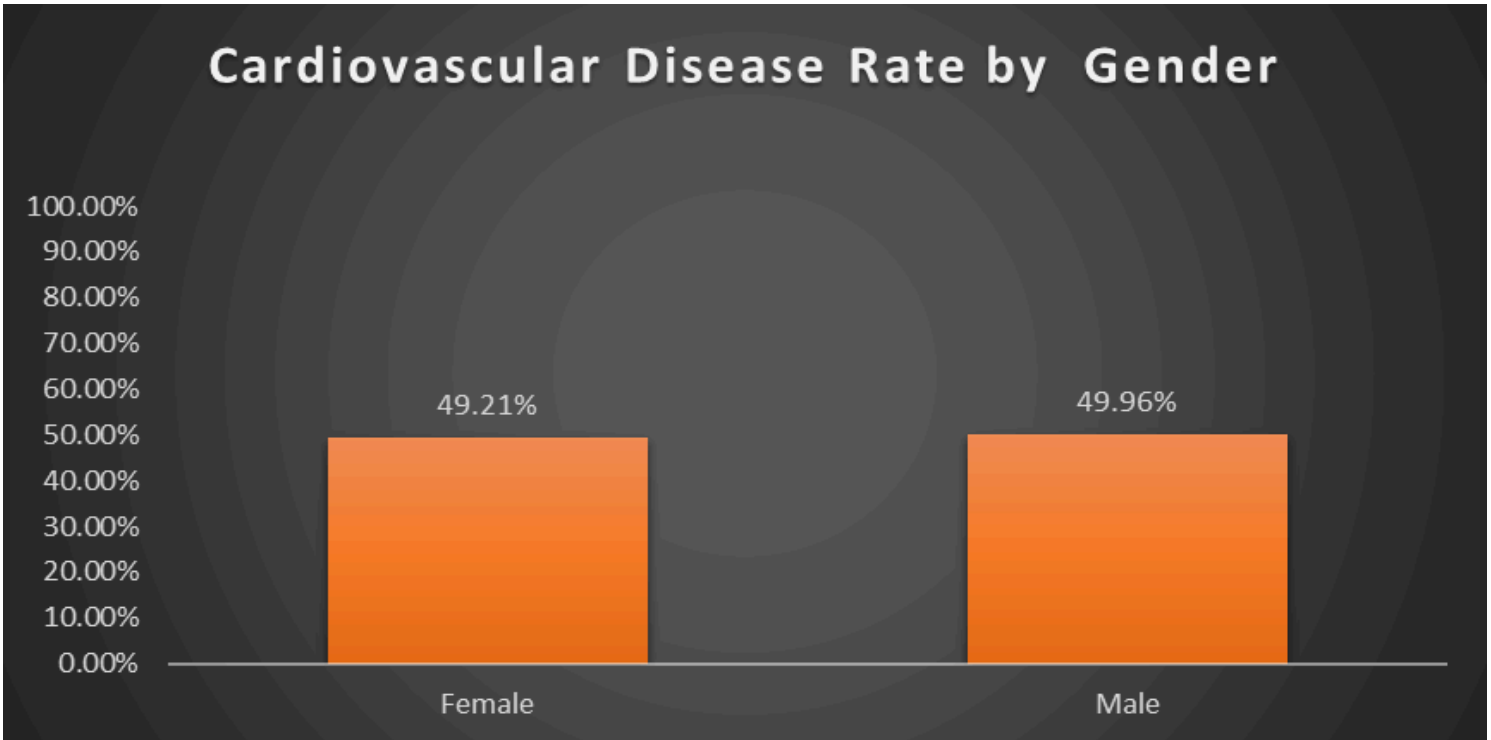
Cardiovascular disease prevalence is nearly identical across genders, with females at 49.2% and males at 50.0%.

Interpretation:

This minimal difference indicates that gender does not have a significant impact on the likelihood of developing cardiovascular disease in this dataset.

Business/Healthcare Impact:

Since both genders exhibit similar risk levels, preventive healthcare strategies and screening programs should be applied equally to both males and females rather than focusing on one specific group.



KPI 3: Lifestyle Factors vs Cardiovascular Disease

Insight:

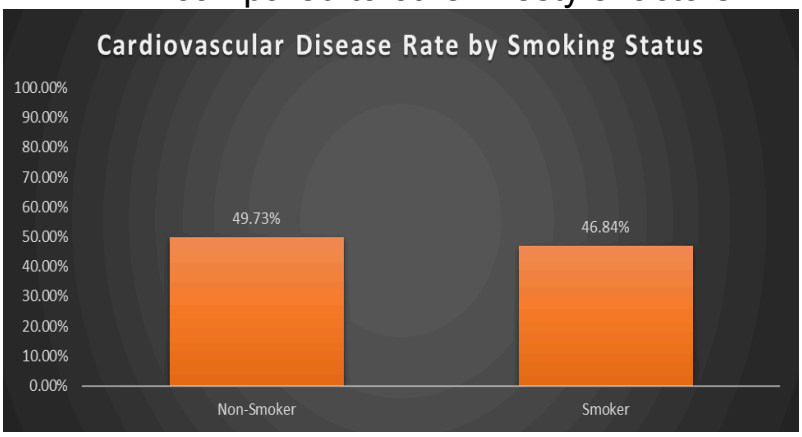
Lifestyle factors such as smoking and alcohol consumption do not show a significant difference in cardiovascular disease prevalence. However, physical activity shows a noticeable impact, with inactive individuals having a higher disease rate (53.3%) compared to active individuals (48.5%).

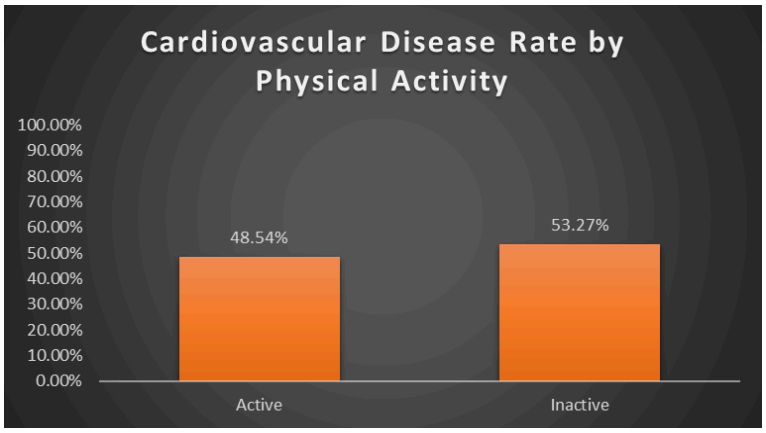
Interpretation:

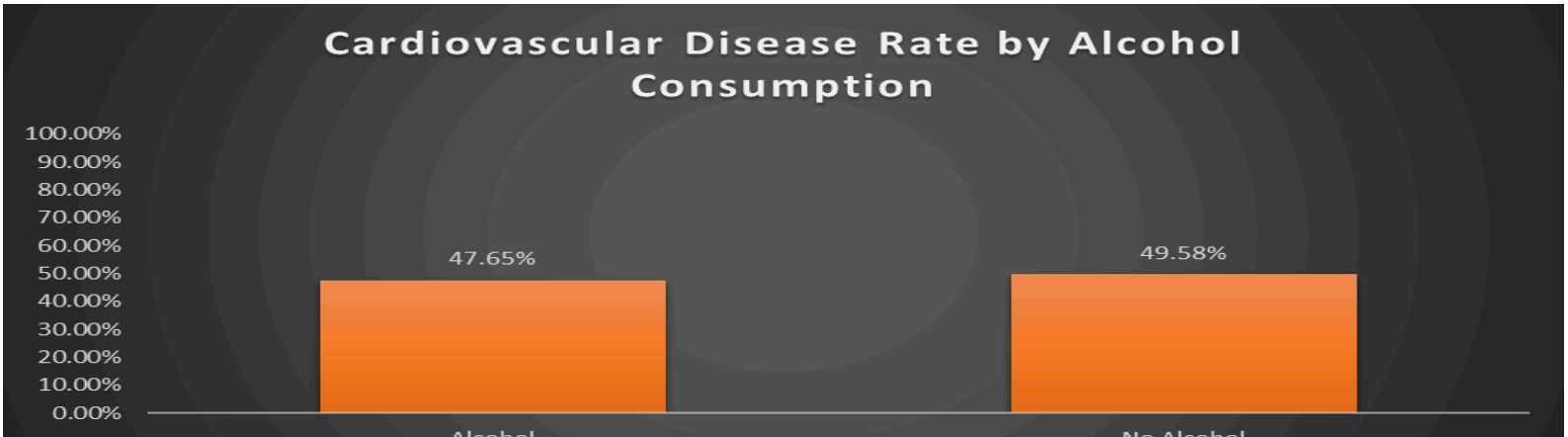
The results suggest that smoking and alcohol consumption alone may not strongly influence cardiovascular disease risk in this dataset. In contrast, physical inactivity appears to be a more relevant factor, indicating that lack of exercise may contribute more significantly to heart disease.

Business/Healthcare Impact:

Promoting physical activity should be a key focus for reducing cardiovascular disease risk. Public health initiatives should prioritize encouraging regular exercise, as it shows a more consistent impact compared to other lifestyle factors.







KPI 4: Combined Impact of Blood Pressure and Cholesterol on Cardiovascular Disease

Insight:

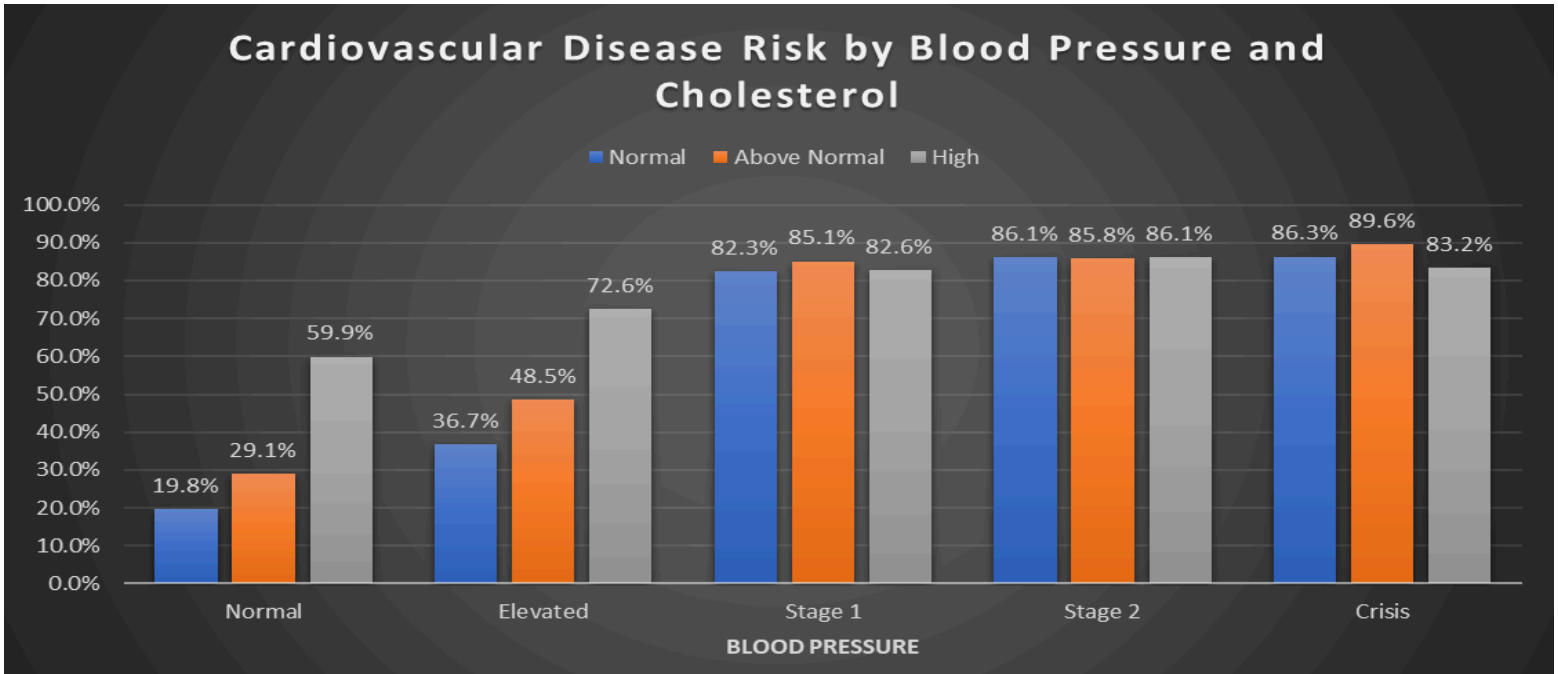
Cardiovascular disease risk increases sharply with rising blood pressure levels. Individuals with Stage 1 or higher blood pressure show disease prevalence above 80%, regardless of cholesterol levels. However, within each blood pressure category, higher cholesterol levels further increase risk. The highest risk (approximately 90%) is observed in individuals with hypertensive crisis and elevated cholesterol.

Interpretation:

Blood pressure acts as the primary driver of cardiovascular disease risk, while cholesterol functions as an amplifying factor. At lower blood pressure levels, cholesterol has a strong differentiating effect on risk. However, once blood pressure reaches Stage 1 or higher, the risk becomes consistently high across all cholesterol levels.

Business/Healthcare Impact:

Effective cardiovascular disease prevention should prioritize blood pressure management as the primary intervention. Additionally, cholesterol control should be emphasized, particularly in individuals with normal or moderately elevated blood pressure, to prevent progression to high-risk categories.



KPI 5: BMI Category vs Cardiovascular Disease

Insight:

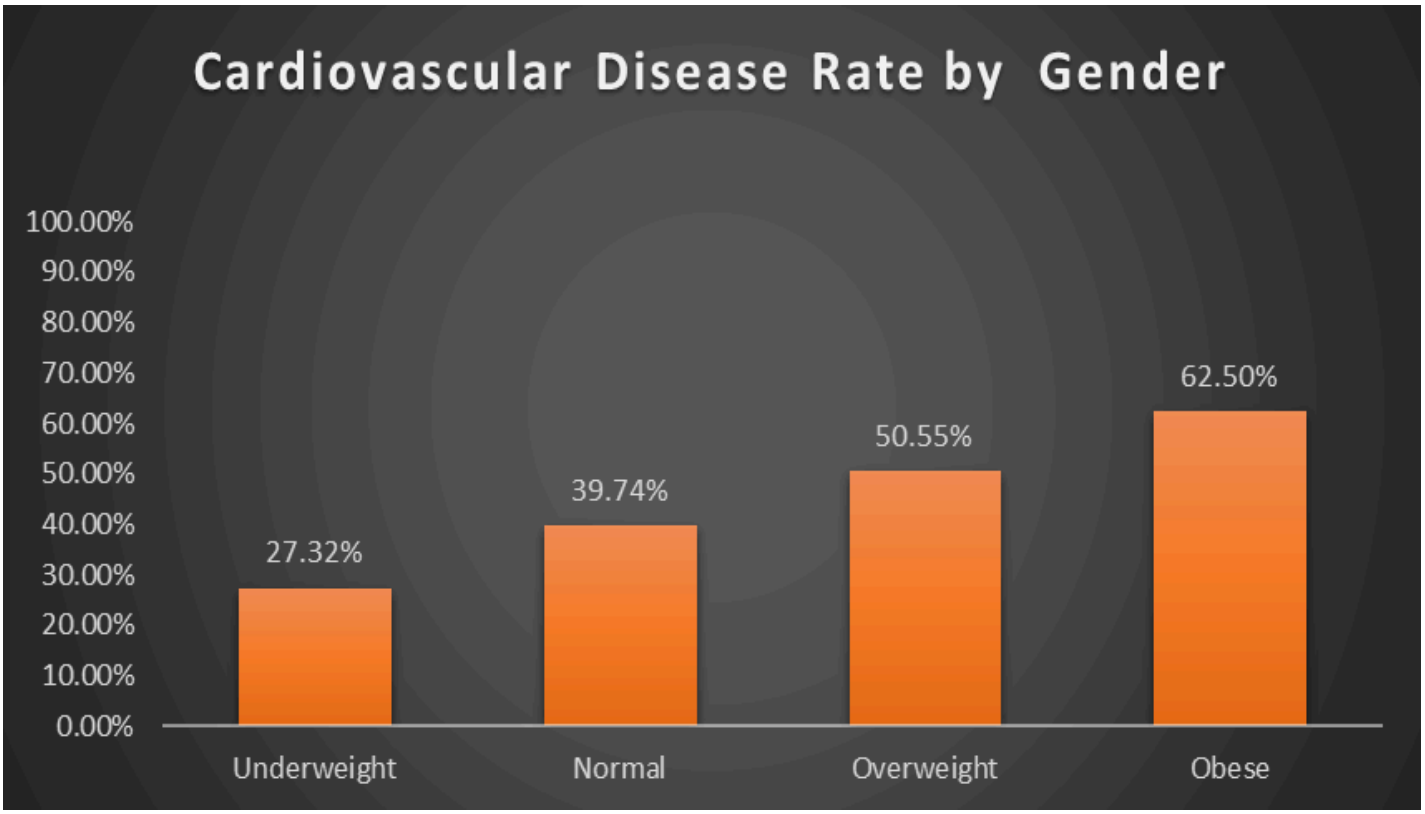
Cardiovascular disease prevalence increases steadily with higher BMI levels, rising from 27.3% in underweight individuals to 62.5% in obese individuals.

Interpretation:

This indicates a strong positive relationship between BMI and cardiovascular disease risk. As body weight increases relative to height, the likelihood of developing heart-related conditions also increases significantly.

Business/Healthcare Impact:

Weight management should be a critical focus area for preventing cardiovascular disease. Encouraging healthy lifestyle habits such as balanced diet and regular exercise can significantly reduce risk, especially for overweight and obese individuals.



KPI 6: High-Risk Patient Segmentation

Insight:

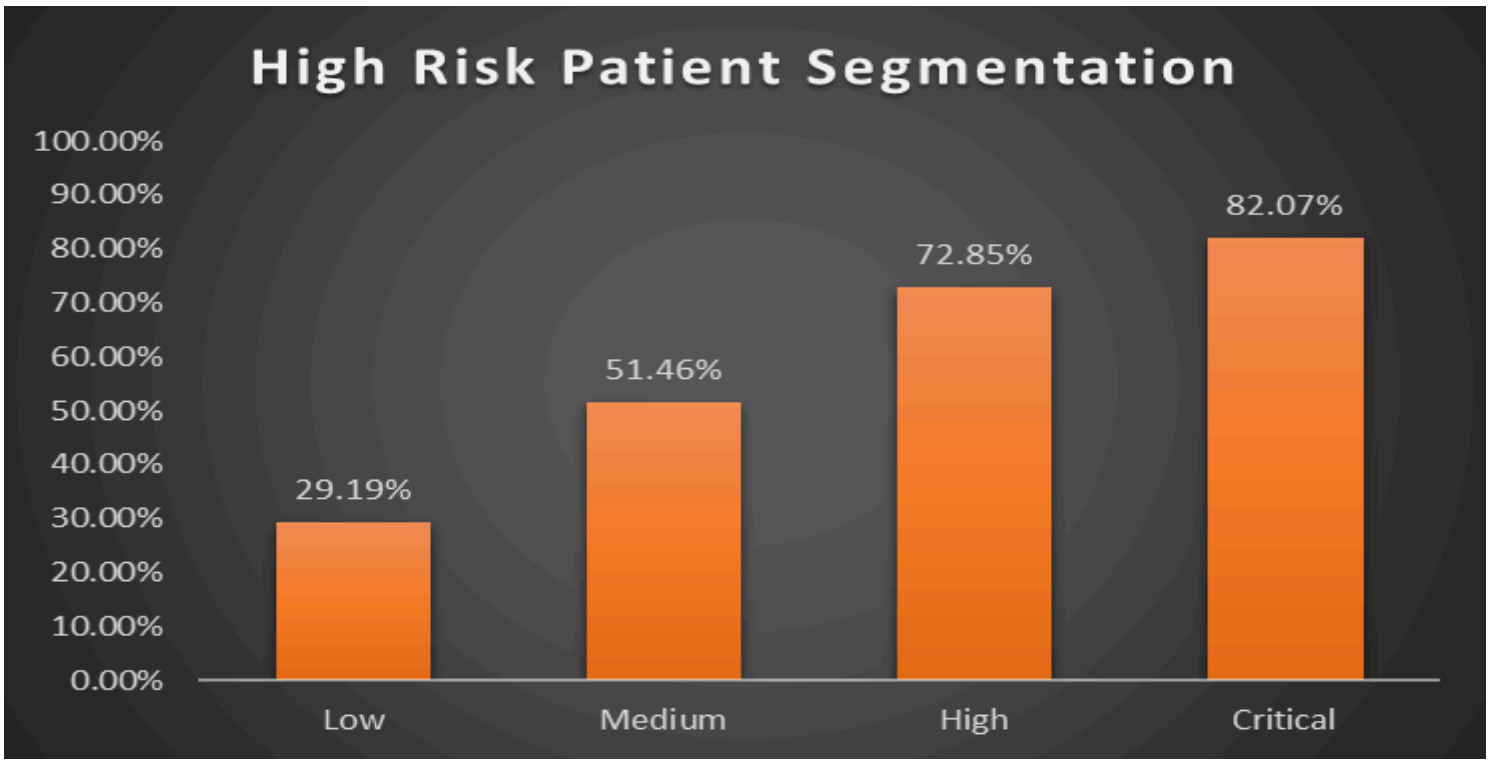
Cardiovascular disease risk increases significantly with higher cumulative risk scores. Individuals in the low-risk group have a disease prevalence of 29.2%, which rises to 51.5% in the medium-risk group, 72.9% in the high-risk group, and reaches 82.1% in the critical-risk group.

Interpretation:

This demonstrates that combining multiple risk factors such as age, blood pressure, cholesterol, BMI, and physical inactivity provides a strong predictive signal for cardiovascular disease. The clear upward trend validates the effectiveness of the risk scoring model.

Business/Healthcare Impact:

The risk segmentation model can be used to identify high-risk patients for early intervention. Healthcare providers can prioritize critical and high-risk individuals for monitoring, preventive care, and targeted treatment strategies.



Final Conclusion: Cardiovascular Disease Risk Analysis

This project analyzed over 70,000 patient records to identify key factors influencing cardiovascular disease. The analysis included data cleaning, feature engineering, KPI-based exploration, and risk segmentation.

The findings show that age and blood pressure are the strongest predictors of cardiovascular disease. Disease prevalence increases significantly with age, particularly in individuals above 50. Blood pressure demonstrated a dominant effect, with risk rising sharply beyond Stage 1 hypertension.

Cholesterol and BMI also showed strong positive relationships with disease risk. Higher cholesterol levels and increased BMI were associated with a substantial rise in cardiovascular disease prevalence. These factors act as important amplifiers of risk, especially when combined with high blood pressure.

Lifestyle factors such as smoking and alcohol consumption did not show a strong individual impact in this dataset. However, physical inactivity showed a moderate but consistent effect, with inactive individuals having higher disease risk.

A key highlight of this project is the development of a rule-based risk scoring model combining age, blood pressure, cholesterol, BMI, and physical activity. The model demonstrated a clear and consistent increase in disease probability across risk segments, rising from 29% in low-risk individuals to over 82% in critical-risk individuals.

Overall, the analysis highlights that clinical factors such as blood pressure, along with BMI and age, play a more significant role in predicting cardiovascular disease compared to individual lifestyle habits. The risk segmentation approach provides a practical framework for identifying high-risk patients and enabling targeted preventive interventions.